

# Synthetic Speech Dataset — Run Report

out\_dir: /content/drive/MyDrive/indic\_synth/out\_deliverable

## 1. Datacard — dataset statistics

| metric                       | value              |
|------------------------------|--------------------|
| utterances checked           | 957                |
| validated (QC passed)        | 922 (96.3%)        |
| distinct languages           | 3                  |
| distinct (language, speaker) | 21                 |
| validated audio              | 3956.4 s (1.099 h) |

### Manifest fields (dataset\_manifest.jsonl)

- **utt\_id** — unique utterance id (e.g. hi\_000000)
- **audio\_filepath** — path to the synthesized wav, relative to out\_dir
- **text** — intended sentence the TTS model was asked to speak
- **speaker\_id** — Kathbath speaker whose voice was cloned
- **gender** — speaker gender (male / female)
- **language** — language code (hi / ml / ta)
- **ref\_id** — reference clip used for voice conditioning
- **ref\_audio\_path** — prepared 24kHz mono reference wav, relative to out\_dir
- **duration** — synthesized audio duration in seconds
- **topic** — sentence-generation topic cell
- **sentence\_type** — declarative / interrogative / imperative / exclamatory
- **status** — pipeline status (validated)
- **qc\_passed** — True if the utterance passed all QC gates
- **qc\_reasons** — list of failed check names ([] when passed)
- **asr\_transcript** — what the ASR model heard (for the CER gate)
- **cer** — character error rate vs intended text (content fidelity)
- **speaker\_sim** — cosine similarity of speaker embeddings synth-vs-ref
- **qc\_checks** — per-check detail: {name, passed, metric, detail}

### Per-language

| language | valid/total utterances | speakers | M/F speakers |
|----------|------------------------|----------|--------------|
| hi       | 300/320                | 7        | 4/3          |
| ml       | 310/318                | 7        | 4/3          |
| ta       | 312/319                | 7        | 4/3          |

### Per-speaker

| language | speaker_id | gender | validated/total utts |
|----------|------------|--------|----------------------|
| hi       | 38         | male   | 42/44                |

|    |      |        |       |
|----|------|--------|-------|
| hi | 164  | female | 41/48 |
| hi | 186  | male   | 45/48 |
| hi | 229  | female | 43/44 |
| hi | 934  | male   | 44/44 |
| hi | 939  | female | 41/44 |
| hi | 1179 | male   | 44/48 |
| ml | 42   | male   | 44/46 |
| ml | 474  | female | 43/44 |
| ml | 653  | female | 42/44 |
| ml | 802  | female | 43/44 |
| ml | 985  | male   | 43/44 |
| ml | 1132 | male   | 48/48 |
| ml | 1145 | male   | 47/48 |
| ta | 21   | male   | 46/48 |
| ta | 67   | female | 43/44 |
| ta | 154  | female | 48/48 |
| ta | 302  | male   | 46/47 |
| ta | 401  | male   | 44/44 |
| ta | 655  | male   | 43/44 |
| ta | 764  | female | 42/44 |

## Gender balance

| gender | utterances | speakers |
|--------|------------|----------|
| female | 404        | 9        |
| male   | 553        | 12       |

## Distributions — over validated set

| metric                  | min   | avg   | max   |
|-------------------------|-------|-------|-------|
| duration (s)            | 1.66  | 4.29  | 10.48 |
| sentence length (chars) | 30    | 51.9  | 93    |
| sentence length (words) | 5     | 7.1   | 14    |
| cer                     | 0.000 | 0.020 | 0.140 |
| speaker_sim             | 0.601 | 0.789 | 0.926 |

## 2. Quality report

### QC yield (audio gates)

| metric    | value  |
|-----------|--------|
| checked   | 957    |
| passed    | 922    |
| failed    | 35     |
| pass rate | 0.9634 |

## QC checks run

- **silence** — rejects near-silent / mostly-inactive audio
- **clipping** — rejects audio with excessive clipped samples
- **truncation** — rejects clips that look cut off (loud tail energy)
- **dur\_per\_char** — duration-to-char ratio must fall in [0.04, 0.30] s/char
- **looping\_audio** — rejects repeated/looping audio segments
- **cer** — content fidelity — ASR CER vs intended text (gate < 0.15)
- **looping\_text** — rejects repeated tokens in the ASR transcript
- **speaker\_cos** — speaker fidelity — cosine sim synth-vs-ref (gate >= 0.60)

## QC rejection breakdown

| failed check | utterances | % of failures |
|--------------|------------|---------------|
| speaker_cos  | 20         | 57.1%         |
| cer          | 11         | 31.4%         |
| truncation   | 5          | 14.3%         |

35 utterances failed; a single utterance can trip more than one check, so counts may sum above the failure total.

## Per-check metric (min / avg / max, all utterances)

| check         | min     | avg     | max     | # failed |
|---------------|---------|---------|---------|----------|
| silence       | -20.020 | -19.999 | -19.980 | 0        |
| clipping      | 0.000   | 0.000   | 0.000   | 0        |
| truncation    | 0.000   | 0.090   | 0.703   | 5        |
| dur_per_char  | 0.045   | 0.083   | 0.156   | 0        |
| looping_audio | 0.000   | 0.090   | 0.765   | 0        |
| cer           | 0.000   | 0.022   | 0.730   | 11       |
| looping_text  | 1.000   | 1.006   | 2.000   | 0        |
| speaker_cos   | 0.511   | 0.785   | 0.926   | 20       |

## Sentence-generation validation

| metric          | value |
|-----------------|-------|
| sentences seen  | 1160  |
| sentences valid | 957   |

|       |       |
|-------|-------|
| yield | 0.825 |
|-------|-------|

#### Rejection breakdown:

| reason          | count | % of rejections |
|-----------------|-------|-----------------|
| too_short       | 99    | 48.8%           |
| near_duplicate  | 46    | 22.7%           |
| wrong_script    | 41    | 20.2%           |
| exact_duplicate | 17    | 8.4%            |

#### Stage-wise pass / fail

| stage               | kept as     | pass | fail | pass rate |
|---------------------|-------------|------|------|-----------|
| Data Acquisition    | downloaded  | 84   | 0    | 100.0%    |
| Audio Engineering   | prepared    | 84   | 0    | 100.0%    |
| Sentence Generation | valid       | 957  | 203  | 82.5%     |
| TTS Generation      | synthesized | 957  | 0    | 100.0%    |
| Quality Control     | passed      | 922  | 35   | 96.3%     |

### 3. Throughput & cost (T4 GPU)

| stage               | elapsed (s) | process (s) | model load (s) | T4 GPU-hr | items          | throughput           |
|---------------------|-------------|-------------|----------------|-----------|----------------|----------------------|
| Data Acquisition    | 91.9        | 91.9        | 0.0            | 0.0255    | 84 clips       | 54.8 clips/min       |
| Audio Engineering   | 3.1         | 3.1         | 0.0            | 0.0009    | 84 clips       | 1636.4 clips/min     |
| Sentence Generation | 4809.6      | 3900.7      | 908.9          | 1.3360    | 957 sentences  | 11.9 sentences/min   |
| TTS Generation      | 12204.8     | 12165.0     | 39.8           | 3.3902    | 957 utterances | 4.7 utterances/min   |
| Quality Control     | 249.5       | 236.8       | 12.7           | 0.0693    | 957 utterances | 230.1 utterances/min |

#### Totals

| metric                                         | value                 |
|------------------------------------------------|-----------------------|
| total wall-clock                               | 17358.9 s (289.3 min) |
| total compute (T4 GPU-hours)                   | 4.8219                |
| of which model loading                         | 961.4 s (5.5%)        |
| end-to-end yield (validated utts / total time) | 3.19 utts/min         |

All stages ran on a single Colab T4, so total compute = total wall-clock GPU-hours. *elapsed\_sec* is the **actual stage wall-clock** = **model load + processing**; *process\_sec* is the item loop alone and *model load (s)* is the model/resource load. Throughput is computed over the full *elapsed\_sec*, so it reflects real end-to-end cost including weight loading.